

AI Resume Analyzer Meeting (Assignment PDF Conversation)

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Quality Index Improvement: +69 Points (8/100 -> 77/100)

Ambiguity: 80% | Testability: 60% | Completeness: 81%

1. Executive Summary

This report compares software requirements captured from a live stakeholder meeting before and after AI-assisted real-time clarification. By dynamically detecting ambiguous predicates and eliciting precise acceptance criteria during the discussion, requirement quality and testability improved significantly.

2. Stakeholder Clarification Q&A Log

[Accuracy (Relevance Metric)] Q1: What quantitative metric will be used to define 'relevance' for the model output?

Triggered By: "How are we defining relevance?"

Stakeholder Response: Precision@K

[Accuracy (Ranking Algorithm Metrics)] Q2: How should 'relevance' be quantified for the ranking algorithm to ensure objective verification?

Triggered By: "It should rank them based on relevance to the job description."

Stakeholder Response: Weighted combination of skills and experience years

[Requirements Elicitation] Q3: To quantify 'profile strength', which specific metrics or weighted scoring criteria should we use to evaluate candidates?

Triggered By: "Mainly skills and experience. And overall profile strength."

Stakeholder Response: Weighted rubric (e.g., 50% technical, 30% soft skills, 20% experience)

[Candidate Selection Criteria] Q4: What specific technical competencies or performance indicators define a 'strong' candidate versus an 'average' one for this role?

Triggered By: "Yes, but not strictly. Sometimes a strong fresher is better than someone with 5 average years."

Stakeholder Response: Defined technical assessment score (e.g., >80% on coding test)

[Accuracy and Fairness] Q5: What are the specific precision and recall targets for the shortlisting algorithm, and what metrics will be used to ensure the model is free from demographic bias?

Triggered By: "We need to build an AI-based resume analyzer that can automatically shortlist candidates for our software engineering roles."

Stakeholder Response: Precision > 80% with human-in-the-loop verification

[Requirements Elicitation] Q6: To ensure we align on expectations, could you define the specific metrics or attributes that qualify a company as 'good', a project as 'solid', and work as 'impactful'?

Triggered By: "Things like good companies, solid projects, impactful work."

Stakeholder Response: Revenue growth/Market position metrics

[Data Engineering] Q7: To define the ingestion requirements, what is the current format of the resumes and decisions, and what level of schema standardization is required for the system?

Triggered By: "We have past resumes and hiring decisions, but they're not very structured."

Stakeholder Response: Structured (SQL/CSV with defined schema)

[Fairness & Bias] Q8: What specific quantitative metric and demographic audit standard should govern bias mitigation across gender and colleges?

Triggered By: "Yes, we must avoid bias, especially related to gender or college background."

Stakeholder Response: Disparate Impact Ratio 0.80 - 1.25 across gender & tier-1/2/3 colleges with PII redaction

[Acceptance Criteria] Q9: What specific metrics or validation results are required for HR to formally sign off on the system's output?

Triggered By: "It should be good enough so that HR trusts it."

Stakeholder Response: Manual audit pass rate of >98% over a 30-day trial

[Performance] Q10: What is the required end-to-end response time / latency target for processing a single resume and batch uploads?

Triggered By: "Ideally quick."

Stakeholder Response: Single resume < 1.5s (p95); Batch of 100 < 30s

[Project Scope] Q11: What is the target MVP delivery timeframe and essential scope boundary for initial release?

Triggered By: "We need an MVP soon."

Stakeholder Response: 2-week prototype with basic resume text extraction and skill keyword matching

[VAGUENESS] Q12: To ensure this feature meets your needs, how would you define 'useful' in terms of measurable outcomes or specific tasks it must enable?

Triggered By: "Yes, that would be useful."

Stakeholder Response: Provides real-time reporting dashboard

[Performance (Latency)] Q13: To define the performance requirement, what is the maximum acceptable response time (latency) for this operation under typical load?

Triggered By: "It shouldn't be slow."

Stakeholder Response: < 200ms (Real-time)

3. Refined Functional Requirements (FR)

FR-01: Multi-Format Resume Ingestion & Parsing [Priority: Must Have]

Description: The system shall ingest resumes in PDF and DOCX formats (up to 10MB per file) and parse text into structured JSON according to stakeholder specification: Structured (SQL/CSV with defined schema).

Acceptance Criteria:

- Given a valid PDF or DOCX resume under 10MB, When uploaded, Then the system extracts text and generates valid structured JSON.
- Given a corrupted or unsupported file type, When uploaded, Then the system returns an informative HTTP 422 error.

Verification Method: Automated Integration Test & JSON Schema Validation

FR-02: Deterministic Multi-Factor Candidate Scoring Engine [Priority: Must Have]

Description: The system shall calculate composite candidate relevance scores using stakeholder-defined weighting: Weighted combination of skills and experience years.

Acceptance Criteria:

- Given a job description and candidate profile, When scored, Then the final score strictly follows the stakeholder weighting formula.
- Given an exceptional fresher with high project complexity, When ranked against generic experience, Then the fresher outranks when composite score is higher.

Verification Method: Unit Test & Ranking Algorithm Regression Suite

FR-03: Model Explainability (Pending Output Schema) [Priority: Medium]

Description: The system shall provide explainability for recommendations (presentation format pending stakeholder clarification).

Acceptance Criteria:

- Explainability feature available

FR-04: Fairness Audit & Demographic Parity Dashboard [Priority: Must Have]

Description: The system shall execute automated Disparate Impact Ratio audits across gender and college tiers complying with: Precision > 80% with human-in-the-loop verification.

Acceptance Criteria:

- Given a candidate batch, When analyzed, Then names, gender indicators, and college brand names are redacted prior to feature extraction.
- Given batch recommendations, When DIR is below 0.80 or above 1.25, Then the system alerts the compliance team.

Verification Method: Automated Bias Test Suite on Synthetic Demographic Cohorts

4. Categorized Non-Functional Requirements (NFR)

NFR-PERF-01: Parsing & Ranking Latency SLO (Performance) [Priority: Critical]

Description: The system shall meet stakeholder-defined latency SLOs: Single resume < 1.5s (p95); Batch of 100 < 30s.

Measurable Metric / Target: Single resume < 1.5s (p95); Batch of 100 < 30s

Verification Method: Automated Performance Benchmark

NFR-FAIR-01: Algorithmic Fairness & Bias Mitigation (Fairness & Bias) [Priority: Critical]

Description: The system shall enforce bias mitigation in accordance with stakeholder parameters: Precision > 80% with human-in-the-loop verification.

Measurable Metric / Target: Precision > 80% with human-in-the-loop verification

Verification Method: Automated Fairness Compliance Audit Suite

NFR-ACC-01: Shortlisting Precision & Ranking Quality (Accuracy & Quality) [Priority: Critical]

Description: The candidate shortlisting model shall achieve stakeholder-defined accuracy targets: Precision@K.

Measurable Metric / Target: Precision@K

Verification Method: Automated Model Evaluation Pipeline on Test Set

NFR-EXP-01: Model Explainability (Unresolved) (Explainability) [Priority: Medium]

Description: Explainability format and latency SLO pending stakeholder input.

Measurable Metric / Target: Unspecified - Awaiting Stakeholder Clarification

NFR-SEC-01: Data Privacy, PII Protection & Storage Security (Security & Privacy) [Priority:

Critical]

Description: All candidate resumes and personal data must be encrypted at rest using AES-256 and in transit using TLS 1.3, with RBAC compliant with GDPR/CCPA based on data pipeline: Structured (SQL/CSV with defined schema).

Measurable Metric / Target: AES-256 at rest, TLS 1.3 in transit, 100% audit logging

Verification Method: Automated SAST/DAST Security Scan

NFR-SCOPE-01: MVP Milestone Delivery & Scope Boundaries (Project Scope) [Priority: High]

Description: The core MVP deliverable shall be completed within stakeholder-specified timeframe: 2-week prototype with basic resume text extraction and skill keyword matching.

Measurable Metric / Target: 2-week prototype with basic resume text extraction and skill keyword matching

Verification Method: Sprint Review & User Acceptance Testing

5. Quality Evaluation & Comparison Audit

Evaluation Dimension	Without		With	
	Clarification (Raw)		Clarification (Refined)	Improvement
Overall Quality Index	8/100 (Poor)		77/100 (Good)	+69 pts
Ambiguity (Lower is better)	100%		20%	+80%
Completeness Coverage	0%		81%	+81%
Testability / Verifiability	0%		60%	+60%
Specificity & Measurability	0%		80%	+80%
Traceability & Structure	50%		90%	+40%

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